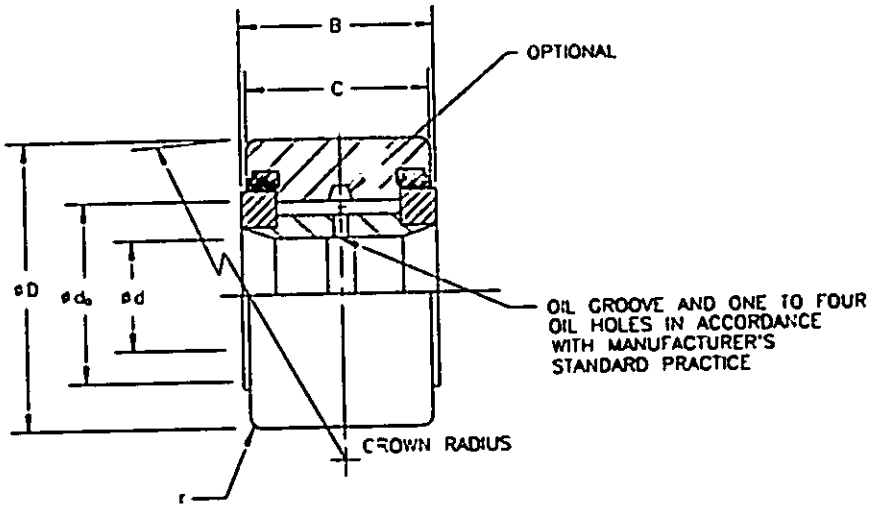
THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION MIL-B-3990

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

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(B) TABLE I

DIMENSIONS IN INCHES

DASH NO	ød BORE	øD OUTSIDE DIAMETER	C OUTER RING WIDTH	B OVERALL WIDTH	øDo CLAMPING DIAMETER MINIMUM	CROWN RADIUS REF	r MAX FILLET CORNER WILL CLEAR	TOTAL RADIAL CLEARANCE MAX	CAPACITY AS A TRACK ROLLER	TRACK CAPACITY LBS	LIMIT LOAD RATING LBF	MASS (APPROX) LB
-121	0.2500	0.7500	0.500	0.5625	0.500	10.00	0.016	0.0023	2000	1030		0.06
-141	0.2500	0.8750	0.500	0.5625	0.500	10.00	0.016	0.0023	2000	1160	2600	0.08
-161	0.3125	1.0000	0.625	0.6875	0.641	12.00	0.032	0.0023	2900	1480	3900	0.15
-181	0.3125	1.1250	0.625	0.6875	0.641	12.00	0.032	0.0023	2900	1670	3900	0.17
-201	0.3750	1.2500	0.750	0.8125	0.766	14.00	0.047	0.0023	4500	2330	5500	0.24
-221	0.3750	1.3750	0.750	0.8125	0.766	14.00	0.047	0.0023	4500	2570	5500	0.30
-241	0.4375	1.5000	0.875	0.9375	0.891	20.00	0.063	0.0023	5800	3380	7300	0.41
-261	0.4375	1.6250	0.875	0.9375	0.891	20.00	0.063	0.0023	5800	3660	7300	0.50
-281	0.5000	1.7500	1.000	1.0625	1.047	20.00	0.063	0.0023	8100	4500	10300	0.64
-301	0.5000	1.8750	1.000	1.0625	1.047	20.00	0.063	0.0023	8100	4820	10300	0.80
-321	0.6250	2.0000	1.250	1.3125	1.203	24.00	0.094	0.0023	11100	6550	13800	1.05
-361	0.6250	2.2500	1.250	1.3125	1.203	24.00	0.094	0.0023	11100	7370	13800	1.32
-401	0.7500	2.5000	1.500	1.5625	1.313	30.00	0.094	0.0023	16200	9250	21600	1.80
-441	0.7500	2.7500	1.500	1.5625	1.313	30.00	0.094	0.0023	16200	10200	21600	2.25
-481	1.0000	3.0000	1.750	1.8125	1.750	30.00	0.125	0.0023	24600	13400	32800	3.10
-521	1.0000	3.2500	1.750	1.8125	1.750	30.00	0.125	0.0023	24600	14500	32800	3.62
-561	1.1250	3.5000	2.000	2.0625	1.922	30.00	0.125	0.0023	31200	18300	41700	4.95
-641	1.2500	4.0000	2.250	2.3125	2.281	30.00	0.125	0.0023	41000	24100	54700	7.05

(B) DENOTES CHANGE

INCH-POUND

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AT NAVY- AIR FORCE-99 DLA- REVIEW: 11, 99, AS, AT, IS USER: PROJECT NUMBER: 3110-0853	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, TRACK ROLLER, SEALED, TYPE X, ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21442 05 DEC 94 REV B SUPERSEDING MS21442A 31 MAR 81 AMSC- N/A FSC 3110
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(B) TABLE II TOLERANCE LIMITS

d BASIC BORE		ALLOWABLE DEVIATION FROM d OF SINGLE MEAN DIA., dmp		ALLOWABLE DEVIATION FROM OVER-ALL WIDTH B	
OVER	INCL	HIGH	LOW	HIGH	LOW
0.1875	0.7500	+0.0002	−0.0004	+0.005	−0.010
0.7500	1.2500	+0.0001	−0.0005	+0.005	−0.010

D BASIC OUTSIDE DIAMETER		ALLOWABLE DEVIATION FROM D OF SINGLE MEAN DIA., dmp		ALLOWABLE DEVIATION FROM OVER RING WIDTH C	
OVER	INCL	HIGH	LOW	HIGH	LOW
0.625	4.000	0.0020	−0.0000	+0.005	−0.005

REQUIREMENTS:

1. MATERIAL: STEEL, MIL-S-7493, MIL-S-8690, ASTMA 304 A322/A331, QQ-S-700, FED. STD. 66, AISI/SAE STEEL NO. 50100, 51100, AND 52100. SEALS AND BACKING RINGS-ACETAL RESIN PER L-P-392 OR NYLON PER L-P-410, TYPE 6/6, WEAR RESISTANT GRADE, ELASTOMERIC POLYESTER. CONSTRUCTION OPTIONAL.

2. FINISH:

   A. OUTER RING-OUTER RING CHROME PLATED PER QQ-C-320, CLASS 2, O.D. AND O.D. CORNERS .0004-.0007 IN. THICKNESS. FACES MINIMUM .0003 IN THICKNESS.

   (B) B. INNER RING -ALL EXTERNAL SURFACES EXCEPT THE INNER RING BORE, ZINC-NICKEL PLATED IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE I, CLASS 2, WITH .0003 TO .0006 THICKNESS. ALTERNATE: ALL EXPOSED SURFACES HAVE BLACK OXIDE FINISH PER MIL-C-13924.

   (B) C. ENDPLATES-ALL EXPOSED SURFACES ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE I, CLASS 2, WITH A THICKNESS OF .0003 TO .0006 INCHES.

3. LUBRICATION: MIL-G-81322 OR MIL-G-23827. ALL BEARINGS SHALL BE PREPACKED WITH GREASE CONFORMING TO MIL-G-81322 UNLESS OTHERWISE SPECIFIED. IF MIL-G-23827 IS REQUIRED ADD THE LETTER G AFTER THE MS PART NUMBER.

4. MARKING: THE MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.

5. PART NUMBER: THE MS PART NUMBER SHALL CONSIST OF THE MS NUMBER PLUS THE DASH NUMBER. EXAMPLE: MS21442-121 OR MS21442-121G.

(B) 6. PACKAGING: BEARING SHALL BE INDIVIDUALLY PACKAGED TO THE REQUIREMENTS OF MIL-P-197. PACKAGE MARKED WITH MANUFACTURER'S NAME OR TRADEMARK, AND DATE OF LUBRICATIUON BY MONTH AND YEAR.

7. DIMENSIONS: ALL DIMENSIONS ARE AFTER PLATING AND ARE IN INCHES.

8. TEMPERATURE: BEARING OPERATING TEMPERATURE RANGE −65°F TO 250°F.

9. LIMIT LOAD: TO FULLY REALIZE LIMIT LOAD RATING, COMPENSATION MUST BE PROVIDED FOR STUD DEFLECTION IN ORDER TO ASSURE OPTIMIM TRACK CONTACT UNDER LOAD. REMOVE ALL BURRS AND SHARP EDGES.

NOTES:

1. DESIGN FEATURES: FOR DESUGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER ANY OTHER DOCUMENTS REFERENCED HEREIN.

2. REFERENCED DOCUMENTS: SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY-AT NAVY- AIR FORCE-99 DLA- REVIEW: 11, 99, AS, AT, IS USER: PROJECT NUMBER: 3110-0853	MILITARY SPECIFICATION SHEET TITLE BEARING, ROLLER, NEEDLE, TRACK ROLLER, SEALED, TYPE X, ANTIFRICTION, INCH	SPECIFICATION SHEET NUMBER MS21442 05 DEC 94 REV B SUPERSEDING MS21442A 31 MAR 81 AMSC- N/A FSC 3110
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